

Harbor Point Microgrid: Operations Brief

Fictional document written for the Flashcard Maker demo. Harbor Point, its operator and every figure below are invented.

1. Site and System Overview

Harbor Point is a waterfront district microgrid that serves a ferry terminal, a cold-storage warehouse and a residential block of fourteen buildings. It can run connected to the regional utility or as an island when the utility supply is lost.

Generation comes from 2.4 MW of rooftop and canopy solar arrays. Storage is a 4 MWh lithium iron phosphate battery housed in two containers beside the warehouse. Two 800 kW biodiesel generators provide backup when solar output and stored energy cannot cover the critical load.

All of these resources meet the utility at a single point of common coupling, where a motorised breaker and a protection relay decide whether the district is connected or islanded.

1.1 Control and Monitoring

A central microgrid controller schedules every resource. It runs an economic dispatch cycle every five minutes, using the latest solar forecast, the battery state of charge and the measured load.

Faster decisions are left to local devices. Protection relays and the battery inverter react within milliseconds, while the controller only sets their targets. Operating data is recorded by a SCADA historian at one-second resolution and retained for three years.

1.2 Load Tiers

Loads are grouped into three tiers so that the most important equipment is protected first. Tier 1 covers the ferry terminal emergency systems and the cold-storage compressors. Tier 2 covers the residential block's lifts, water pumps and corridor lighting. Tier 3 covers everything else, including electric-vehicle chargers and decorative lighting.

2. Storage and Dispatch

The battery is the microgrid's most flexible resource, so its limits are set conservatively. While grid-connected, the battery is kept between 20% and 90% state of charge to slow cell ageing.

During storm season, from 1 June to 30 September, the controller holds a reserve of at least 35% state of charge so that the district can island at short notice.

Peak shaving is the battery's main grid-connected task. When the district's import from the utility rises above 1.8 MW, the battery discharges to hold the import at that level.

2.1 Dispatch Order

When the district must serve its own load, the controller uses solar output before anything else. Stored energy covers the remaining shortfall, and the biodiesel generators are started only when the battery falls below 25% state of charge during an island.

The generators then run at a fixed set point and recharge the battery until it reaches 60%, after which they shut down to save fuel.

2.2 Battery Limits

The battery's measured round-trip efficiency is 88%. To protect its warranty, the controller limits use to one full equivalent cycle per day.

3. Islanding and Fault Response

The protection relay at the point of common coupling watches the utility supply continuously. It opens the breaker and islands the district if utility voltage stays outside 0.88 to 1.10 per unit for more than two seconds, or if frequency leaves the 59.3 to 60.5 Hz band.

When the breaker opens, the battery inverter switches to grid-forming mode within 200 milliseconds and sets the island's voltage and frequency.

If the available supply cannot carry the whole load, Tier 3 loads are shed first, then Tier 2. Tier 1 loads are never shed automatically.

Reconnection is never automatic. The controller resynchronises with the utility only after five minutes of stable utility voltage and an operator's confirmation.

4. Maintenance and Reporting

Each month, technicians run a thermal scan of every battery rack and record any cell that runs five degrees warmer than its neighbours.

Each quarter, both generators are tested on a load bank at 75% of rated output for two hours.

Once a year, the operator carries out a full islanding test, witnessed by the utility, and files the results with the regional regulator.

Any unplanned island, shed load or equipment fault must be written up as an incident report in the operations log within 48 hours.

The operator's headline performance target is 99.95% availability for Tier 1 loads, measured over a rolling twelve months.